

3.1 To keep units consistent, start by converting to micrometers, $x = 100\text{nm} = 0.1\mu\text{m}$.

First, calculating values of D for wet and dry oxygen processes, assuming 100°C and $1000^\circ\text{C} = 1273.15^\circ\text{K}$, using $D = D_0 e^{\frac{-E}{kT}}$

$$D_{\text{wet}} = (9.7 \times 10^7) e^{\frac{2.05}{0.1097}} = 0.7430 \text{ (linear)} = \frac{B}{A}$$

$$D_{\text{wet}} = 386 e^{\frac{0.78}{0.1097}} = 0.3152 \text{ (parabolic)} = B$$

$$D_{\text{dry}} = (3.71 \times 10^6) e^{\frac{2.00}{0.1097}} = 0.0448 \text{ (linear)} = \frac{B}{A}$$

$$D_{\text{dry}} = 772 e^{\frac{1.23}{0.1097}} = 0.0104 \text{ (parabolic)} = B$$

Then, we will use the formula

$$X_o(t) = 0.5A \left[\left\{ 1 + \frac{4B}{A^2}(t + \tau) \right\}^{\frac{1}{2}} - 1 \right]$$

and solve for t

For wet O_2 , we assume an initial oxide layer thickness $X_i = 0 \text{ nm}$, so that $\tau = 0$. Solving for a final thickness of $0.1\mu\text{m}$,

$$0.1 = 0.2121 \left[\left\{ 1 + \frac{1.2608}{0.1799}(t) \right\}^{\frac{1}{2}} - 1 \right]$$

$$t = \left(\left[\frac{0.1}{0.2121} + 1 \right]^2 - 1 \right) \left(\frac{0.1799}{1.2608} \right) = 0.1663 \text{ hrs} = 9.98 \text{ mins.}$$

For dry O_2 , we assume an initial oxide layer thickness $X_i = 25 \text{ nm}$, so that $\tau = \frac{0.025^2}{0.0104} + \frac{0.025}{0.0448} = 0.6181$. Solving again for a final thickness of $0.1\mu\text{m}$,

$$t = \left(\left[\frac{0.1}{0.1161} + 1 \right]^2 - 1 \right) \left(\frac{0.0539}{0.0416} \right) - 0.6181 = 3.58 \text{ hrs.}$$

In conclusion, at $T = 1000^\circ\text{C}$, wet oxidation is preferable for the much faster growth rate up to 100nm .

3.3 Given the differential equation

$$\frac{dX_o}{dt} = \frac{J}{M},$$

where J is the oxidizing flux, given as $J = \frac{DN_0}{(X_o + \frac{D}{k_s})}$, and M is the number of molecules of the oxidizing species incorporated into a unit volume of the resulting oxide, then

$$\frac{dX_o}{dt} = \frac{DN_0/M}{(X_o + D/k_s)}.$$

As shown in the textbook, $A = \frac{2D}{k_s}$ and $B = \frac{2DN_0}{M}$, and so the equation can be reduced to

$$\frac{dX_o}{dt} = \frac{\frac{B}{2}}{X_o + \frac{A}{2}}.$$

Using separation of variables,

$$\frac{B}{2} dt = \left(X_o + \frac{A}{2} \right) dX_o.$$

Integrating both sides,

$$\int \frac{B}{2} dt = \int \left(X_o + \frac{A}{2} \right) dX_o$$

$$= \frac{Bt}{2} = \frac{1}{2} X_o^2 + \frac{AX_o}{2}.$$

Solving for t ,

$$t = \frac{X_o^2}{B} + \frac{X_o}{\frac{B}{A}}.$$

3.6 (a) Estimating the growth time using Fig. 3.6, $t \approx 7$ hrs.

(b) First finding B and $\frac{B}{A}$ for 1150°C ,

$$D_{\text{wet}} = (9.7 \times 10^7) e^{\frac{2.05}{0.1226}} = 5.3316 \text{ (linear)} = \frac{B}{A}$$

$$D_{\text{wet}} = 386 e^{\frac{0.78}{0.1226}} = 0.6661 \text{ (parabolic)} = B.$$

Using Eq. 3.8,

$$t = \frac{4}{0.6661} + \frac{2}{5.3316} = 6.38 \text{ hrs.}$$

3.13 a) Estimating the time to grow $0.4\mu\text{m}$, Fig. 3.6 shows $\tau \approx 0.45$ hrs and total time to grow $1.4\mu\text{m}$ as $t \approx 3.8$ hrs, then $t - \tau \approx 3.4$ hrs.

b) First, finding kT for 1100°C , $kT = 0.118$. Then, finding B and $\frac{B}{A}$ for wet O_2 ,

$$\frac{B}{A} = (9.7 \times 10^7) e^{\frac{-2.05}{0.118}} = 2.7685$$

$$B = 386 e^{\frac{-0.78}{0.118}} = 0.5198.$$

Starting with an $X_i = 0.4\mu\text{m}$ we may calculate τ as

$$\tau = \frac{X_i^2}{B} + \frac{X_i}{\frac{B}{A}} = \frac{0.4^2}{0.5198} + \frac{0.4}{2.7685} = 0.4523 \text{ hrs.}$$

Then, finding time to grow an additional layer of $X_o = 1\mu\text{m}$,

$$t = \frac{X_o^2}{B} + \frac{X_o}{\frac{B}{A}} - \tau = \frac{1.4^2}{0.5198} + \frac{1.4}{2.7685} - 0.4523 = 3.82 \text{ hrs.}$$

Using Table 3.2, a total film thickness of $1.4\mu\text{m}$ will show an orange hue under daylight fluorescent lighting.

3.14 Using Fig. 3.10 to estimate, a thickness of 70nm is needed to mask a 4-hr boron diffusion at 1150°C , and a thickness of 380nm is needed to mask a 1-hr phosphorus diffusion at 1050°C .